Week 9 : Applications (B+C Chapter 7)

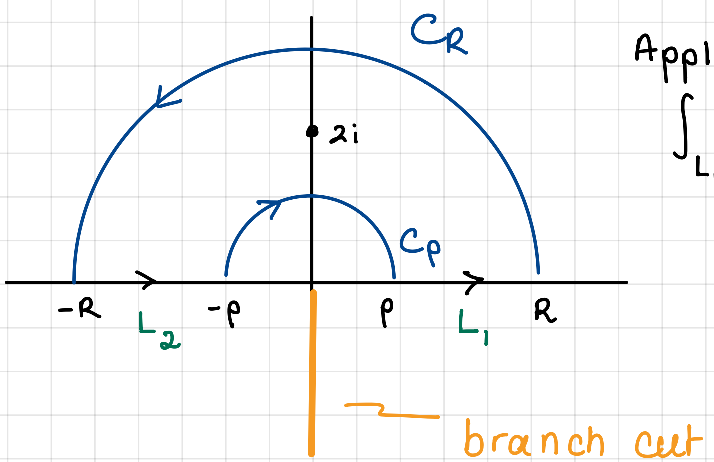
When using residues to compute integrals we may sometimes need to choose a contour to avoid a singularity or branch point.

Example $\displaystyle\int_0^\infty \frac{\ln x}{(x^2+4)^2}\,dx$

Consider $f(z) = \dfrac{\log z}{(z^2+4)^2}$ where $|z|>0$ and $-\frac{\pi}{2} < \arg z < \frac{3\pi}{2}$.

f is analytic on the plane except a branch point at zero, a pole at $z=2i$ and a branch cut on $(0, \infty i)$.

So we consider the following simple closed contour.



Applying the Cauchy Residue theorem:

$$\int_{L_1} f(z)\,dz + \int_{C_R} f(z)\,dz + \int_{L_2} f(z)\,dz + \int_{C_\rho} f(z)\,dz = 2\pi i \operatorname*{Res}_{z=2i} f(z). \qquad (*)$$

Note:
$$\int_{L_2} f(z)\,dz + \int_{L_1} f(z)\,dz = \int_{-R}^{-\rho} \frac{\ln|x| + i\pi}{(x^2+4)^2}\cdot dx + \int_\rho^R \frac{\ln|x|}{(x^2+4)^2}\,dx$$
$$= 2\int_\rho^R \frac{\ln x}{(x^2+4)^2}\,dx + i\pi \int_\rho^R \frac{1}{(x^2+4)^2}\,dx$$

and also $f(z) = \dfrac{\phi(z)}{(z-2i)^2}$ where $\phi(z) = \dfrac{\operatorname{Log} z}{(z+2i)^2}$ is analytic and non-zero.

$$\Rightarrow \operatorname*{Res}_{z=2i} f(z) = \phi'(2i) = \frac{d}{dz}\left(\frac{\operatorname{Log} z}{(z+2i)^2}\right)$$

$$= \left. \frac{(z+2i)^2 \frac{1}{z} - 2(z+2i)\log z}{(z+2i)^4} \right|_{z=2i}$$

$$= \frac{(4i)^2 \cdot \frac{1}{2i} - 2(4i)\log(2i)}{(4i)^4}$$

$$= \frac{8i - 8i\left(\ln 2 + \pi/_2\, i\right)}{256}$$

$$= \frac{\pi}{64} + \frac{1}{32}(1 - \ln 2)i$$

The remaining two terms should go to zero.

- $$\left| \int_{C_R} \frac{\log z}{(z^2+4)^2}\, dz \right| \leq \frac{\ln R + 2\pi}{|R^2 - 4|} \cdot \pi R = \frac{\frac{\ln R}{R} + \frac{2\pi}{R}}{1 - \frac{4}{R^2}} \longrightarrow 0 \quad \text{as } R \to \infty.$$

- $$\left| \int_{C_\rho} \frac{\log z}{(z^2+4)^2}\, dz \right| \leq \frac{-\ln \rho + 2\pi}{|\rho^2 - 4|} \cdot \pi\rho = \frac{-\rho \ln \rho + 2\pi^2 \rho}{|\rho^2 - 4|} \longrightarrow 0 \quad \text{as } \rho \to 0.$$

So taking the limit as $R \to \infty$ and $\rho \to 0$ we get:

$$2\int_0^\infty \frac{\ln x}{(x^2+4)^2}\, dx + i\pi \int_0^\infty \frac{1}{(x^2+4)^2}\, dx = 2\pi i \left(\frac{\pi}{64} + \frac{1}{32}(1 - \ln 2)i \right)$$

Taking the real part: $\boxed{\int_0^\infty \frac{\ln x}{(x^2+4)^2}\, dx = \frac{\pi}{32}(\ln 2 - 1)}$

If we instead take imaginary parts we also get:

$$\int_0^\infty \frac{1}{(x^2+4)^2}\, dx = \frac{\pi}{32}.$$

Definite Integrals of Sines + Cosines

We can use similar techniques to evaluate some definite integrals.

Idea: Suppose we are interested in the integral

$$I = \int_0^{2\pi} F(\sin\theta, \cos\theta)\, d\theta$$

Then we can use $z = e^{i\theta}$ to write

$$\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta}) = \frac{1}{2i}(z - z^{-1}), \quad \cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta}) = \frac{1}{2}(z + z^{-1})$$

and hence

$$I = \int_C F\left(\frac{1}{2i}(z - z^{-1}), \frac{1}{2}(z + z^{-1})\right) \frac{1}{iz}\, dz .$$

where C is the unit circle. We can then use the Cauchy Residue theorem to evaluate the integral.

Example. Compute $\int_0^{2\pi} \frac{d\theta}{1 + a\sin\theta}$ for $-1 < a < 1$ and $a \neq 0$.

Making the substitution from above.

$$\frac{1}{1 + a \cdot \frac{1}{2i}(z - z^{-1})} = \frac{\frac{2i}{a} z}{\frac{2i}{a} z + z^2 - 1}$$

$$\Rightarrow \quad I = \int_0^{2\pi} \frac{d\theta}{1 + a\sin\theta} = \int_C \frac{2/a}{z^2 + \frac{2i}{a} z - 1}\, dz$$

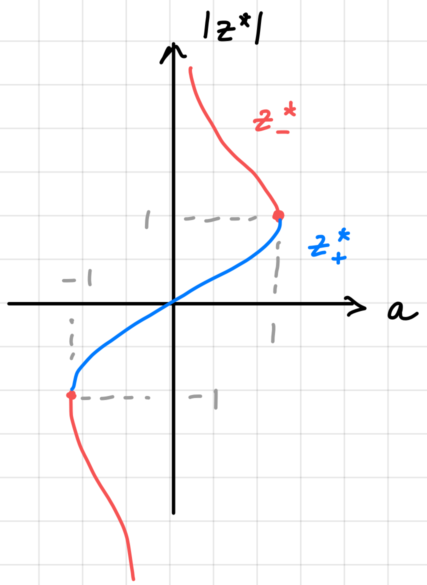
By completing the square:

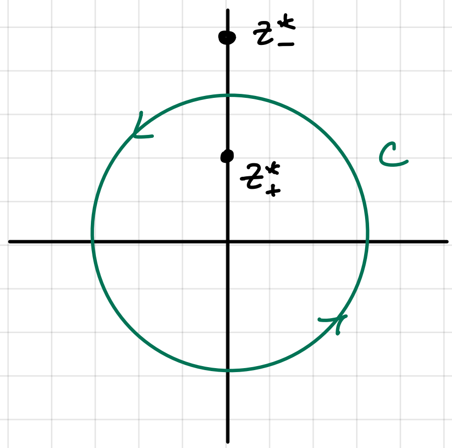
$$z^2 + \frac{2i}{a} z - 1 = z^2 + \frac{2i}{a} z - \frac{1}{a^2} + \frac{1}{a^2} - 1$$
$$= \left(z + \frac{i}{a}\right)^2 + \frac{1}{a^2} - 1$$

and so singularities are found at

$$z_\pm^* = -\frac{i}{a} \pm \sqrt{1 - \frac{1}{a^2}} = -\frac{i}{a} \pm i\sqrt{\frac{1}{a^2} - 1} = \frac{i}{a}\left(-1 \pm \sqrt{1-a^2}\right)$$

We can determine where these points are in the complex plane by plotting $|z^*|$:





Hence,
$$I = \int_C \frac{2/a}{(z - z_+^*)(z - z_-^*)}\, dz = 2\pi i \operatorname*{Res}_{z = z_+^*} f(z) .$$

To compute the residue we note

$$f(z) = \frac{2/a}{z - z_-^*} \cdot \frac{1}{z - z_+^*} =: \frac{\phi(z)}{z - z_+^*}$$

where $\phi(z)$ is analytic and non-zero. Since the pole is simple we find

$$\operatorname*{Res}_{z = z_+^*} f(z) = \phi(z_+^*) = \frac{2/a}{z_+^* - z_-^*} = \frac{2/a}{\frac{i}{a}\left(-1 + \sqrt{1-a^2} - (-1 - \sqrt{1-a^2})\right)}$$

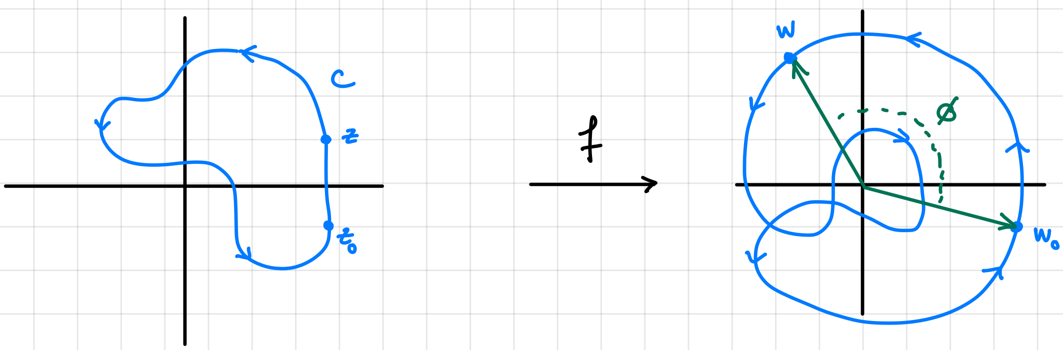
$$= \frac{2}{i\, 2\sqrt{1-a^2}} = -i \frac{1}{\sqrt{1-a^2}}$$

$$\Rightarrow \quad I = 2\pi i \cdot \frac{-i}{\sqrt{1-a^2}} = \frac{2\pi}{\sqrt{1-a^2}}$$

The Arguement Principle

We can also use contour integration to detect zeros and poles of analytic functions.

Suppose f is analytic on the interior of a simple closed contour C. Then the image of C under f, Γ, is also a closed contour.



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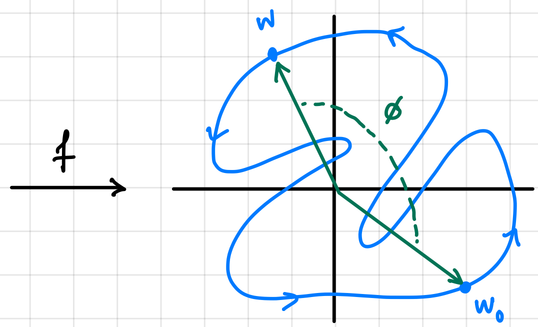


If we also assume f is non-zero on C, then Γ does not pass through 0.

Let $w = f(z)$ and $w_0 = f(z_0)$. We imagine that z is allowed to traverse the path starting from z_0.

Then we track the angle spanned by w and w_0. Once z returns to z_0, w returns to w_0 and so ϕ must be a multiple of 2π.

We define this angle: $\Delta_C \arg f(z) := \phi_1 - \phi_0$

or the winding number of C and f.

It turns out the winding number is determined by the number of roots and poles of f inside C.

Thm: Let C denote a positively oriented simple closed contour and suppose

(a) a function $f(z)$ is analytic inside C except at poles

(b) $f(z)$ is analytic and non-zero on C

(c) counting multiplicities, Z = number of roots of f inside C, P = number of poles of f inside C.

Then $$\Delta_C \arg f(z) = 2\pi(Z-P).$$

Proof: We consider the function $g(z) = \dfrac{f'(z)}{f(z)}$ and a parameterization $z = z(t)$ of C.

Then $$\int_C g(z)\,dz = \int_a^b \frac{f'(z(t))}{f(z(t))} z'(t)\,dt$$

But now, $f(z(t))$ is non-zero and so we can reparameterize this as a path

$$w(t) = f(z(t)) = \rho(t) e^{i\phi(t)}$$

a parameterization for Γ. Then

$$f'(z(t))\,z'(t) = \frac{d}{dt}\big(f(z(t))\big) = \frac{d}{dt}\left(\rho(t)e^{i\phi(t)}\right)$$

$$= \rho'(t)e^{i\phi(t)} + i\rho(t)e^{i\phi(t)}\phi'(t)$$

$$\Rightarrow \int_C \frac{f'(z)}{f(z)}\,dz = \int_a^b \frac{\rho'(t)e^{i\phi(t)} + i\rho(t)e^{i\phi(t)}\phi'(t)}{\rho(t)e^{i\phi(t)}}\,dt$$

$$= \int_a^b \frac{\rho'(t)}{\rho(t)}\,dt + i\int_a^b \phi'(t)\,dt$$

$$= \ln(\rho(t)) + i\,\phi(t)\Big|_{t=a}^{b}$$

But we have a closed curve so $\rho(a) = \rho(b) = |w_0|$

$$\Rightarrow \int_c \frac{f'(z)}{f(z)}\, dz = i\big(\phi(b) - \phi(a)\big) = i\,\Delta_c \arg f(z).$$

On the otherhand we can compute the integral using the Cauchy Residue Theorem. To do this we need to determine what the poles of g are.

Suppose f has a zero of order m at z_0. Then

$$f(z) = (z-z_0)^m h(z) \quad \text{where } h \text{ is analytic and non-zero at } z_0.$$

$$\Rightarrow f'(z) = m(z-z_0)^{m-1} h(z) + (z-z_0)^m h'(z)$$

$$\Rightarrow \frac{f'(z)}{f(z)} = \underbrace{\frac{m}{z-z_0}}_{\text{principal}} + \underbrace{\frac{h'(z)}{h(z)}}_{\text{analytic near } z_0 \text{ as } h(z_0) \neq 0.}$$

Hence, z_0 is a simple pole of g and $\operatorname*{Res}_{z=z_0} g(z) = m$.

Suppose f has a pole of order n at z_0. Then

$$f(z) = \frac{\phi(z)}{(z-z_0)^n} \quad \text{for } \phi(z) \text{ analytic and non-zero at } z_0.$$

$$\Rightarrow f'(z) = \frac{\phi'(z)}{(z-z_0)^n} - \frac{n\,\phi(z)}{(z-z_0)^{n+1}}$$

$$\Rightarrow \frac{f'(z)}{f(z)} = \underbrace{\frac{\phi'(z)}{\phi(z)}}_{\substack{\text{analytic} \\ \text{near } z_0 \text{ as } \phi(z_0)\neq 0.}} - \underbrace{\frac{n}{z-z_0}}_{\text{principal}}$$

Hence, again g has a simple pole and $\operatorname*{Res}_{z=z_0} g(z) = -n$.

So using the Cauchy Residue Theorem:

$$-i\,\Delta_c \arg f(z) = \int_c \frac{f'(z)}{f(z)}\, dz = 2\pi i \sum_k \operatorname*{Res}_{z=z_k} g(z) = 2\pi i(Z - P)$$

$\square$

Rouché's Theorem

Thm: (Rouché)

Let C be a simple closed contour and suppose that

(a) f and g are analytic on and inside C

(b) $|f(z)| > |g(z)|$ for every point $z \in C$.

Then $f(z)$ and $f(z)+g(z)$ have the same number of zeros inside C.

Proof: First note that neither f nor $f+g$ have a zero on C as

$$|f(z)| > |g(z)| \geq 0$$

$$|f(z)+g(z)| \geq ||f(z)| - |g(z)|| > 0 .$$

So using the Arguement Principle we know that

$$Z_f = \frac{1}{2\pi} \Delta_c \arg f(z) \quad , \quad Z_{f+g} = \frac{1}{2\pi} \Delta_c \arg\{f(z)+g(z)\} .$$

are the number of zeros of f and $f+g$ respectively.

But by using additivity of the arguement:

$$\Delta_c \arg\{f(z)+g(z)\} = \Delta_c \arg\left\{f(z)\left(1+\frac{g(z)}{f(z)}\right)\right\}$$

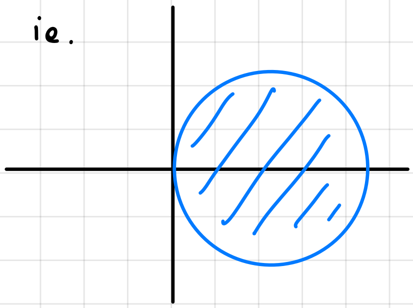
$$= \Delta_c \arg f(z) + \Delta_c \arg\left(1+\frac{g(z)}{f(z)}\right)$$

$$\Rightarrow \quad Z_{f+g} = Z_f + \frac{1}{2\pi} \Delta_c \arg\left(1+\frac{g(z)}{f(z)}\right)$$

So if we can show that $F(z) = 1 + \frac{g(z)}{f(z)}$ has winding number 0, we are done.

But note that $|F(z)-1| = \left|\frac{g(z)}{f(z)}\right| < 1$ by assumption.

ie.



The mapping of C through F is contained in the unit ball centered at 1.

The winding number can only be zero as the contour does not surround 0.

$\square$

<u>Example</u>: $z^7 - 4z^3 + z - 1 = 0$ has how many roots inside the unit circle?

Set $g(z) = z^7 + z - 1$ and $f(z) = -4z^3$

Then
$$\left.\begin{aligned} |g(z)| &\leq |z|^7 + |z| + 1 = 3 \\ |f(z)| &= 4|z|^3 = 4 \end{aligned}\right\} \text{ for } z \in \text{unit circle.}$$

We conclude that $f+g$ has the same number of roots inside C as f. But

$$-4z^3 = 0 \iff z^3 = 0 \iff z = 0.$$

So there is exactly 1 root inside C.

<u>Example</u>: From the previous integral example, we can show

$$z^2 + \frac{2i}{a} z - 1 = 0 \qquad (-1 < a < 1 \text{ and } a \neq 0)$$

has a single zero inside the unit circle.

Let $f(z) = \frac{2i}{a} z$ and $g(z) = z^2 - 1$

Then $|f(z)| = \frac{2}{a}$ while $|g(z)| \leq |z|^2 + 1 = 2$.

Since $|a| < 1$ we can apply Rouché's Thm to see that there is a single root in the unit circle.

Inverse Laplace Transforms

The Laplace transform of a function f is defined by

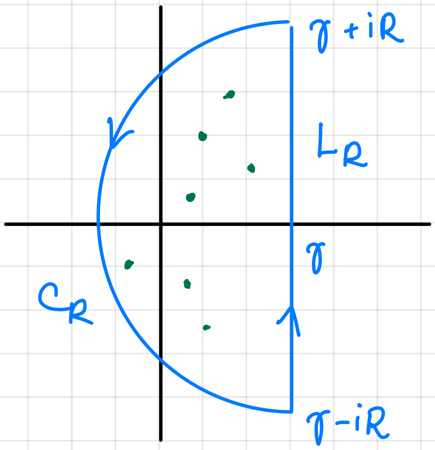
$$F(s) = \int_0^\infty e^{-st} f(t)\, dt$$

Assuming F is analytic for all s except a finite number of isolated singularities, we can find the inverse Laplace transform by the formula

$$f(t) = \frac{1}{2\pi i} \lim_{R\to\infty} \int_{L_R} e^{st} F(s)\, ds$$

$$= \frac{1}{2\pi i}\ \text{P.V.} \int_{\gamma - i\infty}^{\gamma + i\infty} e^{st} F(s)\, ds$$

where L_R is the line segment from $\gamma - iR$ to $\gamma + iR$, and we choose $\gamma \in \mathbb{R}$ large so all the singularities of F lie to the left of the line segment.



This is called the Mellin Inverse Formula, the Bromwich integral or the Fourier-Mellin integral.

Luckily for us, this means we can compute inverse Laplace transform using the Cauchy Residue Theorem.

Indeed by CRT:

$$\int_{L_R} e^{st} F(s)\, ds = 2\pi i \sum_{k=1}^{N} \operatorname*{Res}_{z=z_k} f(z) - \int_{C_R} e^{st} F(s)\, ds$$

where we can take the limit as $R \to \infty$ to obtain the inverse Laplace transform.

Example: Find the inverse Laplace transform of

$$F(s) = \frac{s}{(s^2+a^2)^2} \qquad (a>0)$$

The singularities of F are $s_0 = ai$ and $\bar{s}_0 = -ai$.

However, unlike in previous examples we are interested in computing the integral of $e^{st}F(s)$ for fixed t.

We can use a nice trick to compute the sum of residues of this function.

Suppose F has a principal part

$$p(s) = \frac{b_1}{s-s_0} + \frac{b_2}{(s-s_0)^2} + \cdots + \frac{b_m}{(s-s_0)^m}$$

then since

$$\begin{aligned} e^{st} &= e^{s_0 t} e^{(s-s_0)t} \\ &= e^{s_0 t}\left(1 + (s-s_0)t + \frac{(s-s_0)^2 t^2}{2!} + \cdots\right) \end{aligned}$$

we know that the principal part of $e^{st}F(s)$ is found by from the singular terms of

$$\begin{aligned} \tilde{p}(s) &= p(s)e^{st} \\ &= \left(\frac{b_1}{s-s_0} + \frac{b_2}{(s-s_0)^2} + \cdots + \frac{b_m}{(s-s_0)^m}\right) e^{s_0 t}\left(1 + (s-s_0)t + \frac{(s-s_0)^2 t^2}{2!} + \cdots\right) \end{aligned}$$

Now the residue is found as the coefficient of the $(s-s_0)^{-1}$ term.

Hence,
$$\operatorname*{Res}_{s=s_0} e^{st}F(s) = e^{s_0 t}\left(b_1 + \frac{b_2}{1!}t + \frac{b_3}{2!}t^2 + \cdots + \frac{b_m}{(m-1)!}t^{m-1}\right).$$

and similarly

$$\operatorname*{Res}_{s=s_0} e^{st}F(s) = e^{\bar{s}_0 t}\left(\bar{b}_1 + \frac{\bar{b}_2}{1!}t + \frac{\bar{b}_3}{2!}t^2 + \cdots + \frac{\bar{b}_m}{(m-1)!}t^{m-1}\right)$$

$$\Rightarrow \operatorname*{Res}_{s=s_0} e^{st}F(s) + \operatorname*{Res}_{s=\bar{s}_0} e^{st}F(s) = 2\,\mathrm{Re}\left\{e^{s_0 t}\left(b_1 + \frac{b_2}{1!}t + \cdots + \frac{b_m t^{m-1}}{(m-1)!}\right)\right\}.$$

In this case: $F(s) = \dfrac{s}{(s-ai)^s(s+ai)^2}$ so expanding at $s = ai$ we find

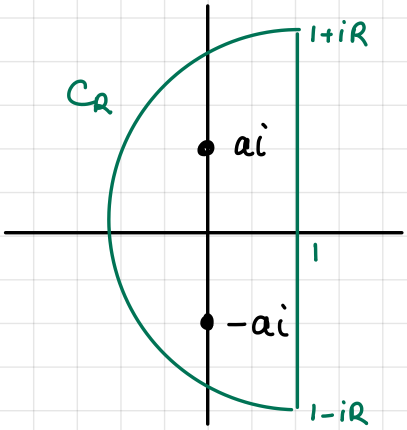
$$p(s) = \frac{\phi(ai)}{(s-ai)^2} + \frac{\phi'(ai)}{(s-ai)} \quad \text{where} \quad \phi(s) = \frac{s}{(s+ai)^2}$$

$$= \frac{-i/4a}{(s-ai)^2} + \frac{0}{s-ai}$$

Hence, $\underset{s=ai}{\text{Res}}\, e^{st}F(s) + \underset{s=-ai}{\text{Res}}\, e^{st}F(s) = 2\,\text{Re}\left[e^{iat}\left(-\frac{i}{4a}t\right)\right]$

$$= \frac{1}{2a} t\sin(at).$$

Using this we now want to use the Cauchy Residue Thm:



$$\int_{L_R} e^{st}F(s)\,ds = \frac{1}{2a}t\sin(at) - \int_{C_R} e^{st}F(s)\,ds$$

If we can show the second integral goes to zero then we have

$$f(t) = \frac{1}{2a}t\sin(at)$$

So we want to bound the modulus.

If $s = 1 + Re^{i\theta}$ then $\left|e^{st}F(s)\right| = \left| e^{(1+Re^{i\theta})t} \dfrac{1+Re^{i\theta}}{((1+Re^{i\theta})^2 + a^2)^2}\right|$

$$\leq \frac{e^{t+Rt\cos\theta}(1+R)}{\left||R^2-1|^2 - a^2\right|^2}$$

$$\Rightarrow \left|\int_{C_R} e^{st}F(s)\,ds\right| \leq \frac{e^{t}(1+R)\pi R}{\left||R^2-1|^2 - a^2\right|^2} \longrightarrow 0 \quad \text{as } R \to \infty$$

since $\cos\theta \leq 0$.